

Burak Demirel

🇪🇺 EU Citizen
☎ +31 6 26123265
✉ burak.demirel@protonmail.com
in demirelbu
🌐 demirelbur

Summary

Machine Learning Engineer with 10+ years of experience building and shipping scalable, production-grade ML systems that operate under strict latency and reliability constraints. Proven track record of translating ambiguous business problems into deployable AI solutions, specializing in reinforcement learning, agentic AI, and LLM-integrated architectures. Experienced in designing distributed training pipelines, optimizing model efficiency, and productionizing models as robust, monitored services in cloud and HPC environments. Strong cross-functional partner who collaborates closely with product, research, and engineering teams to define metrics, iterate on performance, and deliver measurable impact at scale. Fully committed to building my long-term professional future in the Netherlands, where my family have recently relocated.

Professional Experience

Dec 2023 – **Master Researcher in Machine Learning**, *Ericsson AB*, Kista

- Present
- Architected a scalable distributed **multi-agent multi-objective reinforcement learning** platform for 5G coverage and capacity optimisation, enabling large-scale training on an on-premises cloud infrastructure.
 - Developed multi-objective reinforcement learning algorithms to **balance competing network KPIs**, including coverage and capacity, using preference-based optimisation.
 - Built a distributed training pipeline for large-scale experimentation and evaluation of network control policies in simulated 5G environments.
 - Led the design and industrialization of **Agentic AI systems** for autonomous 5G network operations, enabling real-time closed-loop decision-making, by integrating **reinforcement learning (RL)**, **Bayesian optimization**, and **LLM-based reasoning**.
 - Enabled real-time deployment of AI workloads on baseband hardware by identifying latency constraints and applying **policy distillation** to produce compact, resource-efficient models.
 - Accelerated the transition of **AI-native link adaptation** from research prototypes to product development by providing sustained post-handover support, including result interpretation, hyperparameter optimization, and architectural refinement in close collaboration with product units.
 - Built a **PyTorch-native RL library** adopted by ~100 engineers and researchers by integrating multiple internal simulators and standardizing training workflows across teams.
 - Influenced system design and engineering practices across teams by mentoring engineers and serving as a cross-team technical advisor on deployment constraints and long-term architecture.

Jan 2020 – **Senior Researcher in Machine Learning**, *Ericsson AB*, Kista

- Dec 2023
- Established the commercial viability of **AI-native link adaptation** by leading early research and **proof-of-concept development**, demonstrating **spectral efficiency and throughput gains** in close collaboration with product units.
 - Reduced distributed RL training time by **20x** by designing and owning **scalable HPC architectures** with optimized scheduling and high-throughput data pipelines.
 - Architected **high-throughput distributed RL systems** spanning **GPU learners**, **100+ CPU actors** interacting with radio network simulators, and **multi-node coordination** using Redis, ZeroMQ, and PyTorch RPC.
 - Delivered measurable performance improvements in **production-grade 5G RAN systems** by applying reinforcement learning under **real-world latency, reliability, and hardware constraints**.
 - Shaped strategic technical direction by **co-authoring technical papers** and delivering **system-level insights** to senior stakeholders.

- May 2018 – **Development Engineer, Planning & Decision-Making Team, Scania CV AB, Södertälje**
Jan 2020
- Designed robust **automation algorithms** for path tracking and motion planning in **autonomous heavy-duty vehicles**.
 - Drafted and filed multiple **patent applications** in connected transport systems, collaborating with IP legal teams.
 - Supervised a master's thesis project on **real-time automation for motion planning**, aligning academic research with industrial needs.
- Oct 2015 – **Postdoctoral Researcher, Paderborn University, Paderborn**
Mar 2018
- Investigated advanced **control, signal processing and machine learning** techniques for large-scale networked Cyber-Physical Systems.
 - Contributed to grant applications for the German Research Council (DFG) on Cyber-Physical Networking.
 - Taught own graduate course on **Control of Mechatronic Systems** and supervised multiple theses.

Side Projects

Agentic AI Product Recommendation System (E-commerce)

Designed and implemented a conversational recommendation system for product discovery using a multi-agent LLM pipeline. Combined structured SQL filtering over a relational catalog with semantic vector search (ChromaDB + sentence-transformers) to support both precise queries (price, rating, category) and qualitative intent (taste, style, pairing). Ensured factual reliability through schema-validated agent outputs, deterministic rendering of database values, contradiction detection, and automatic retry mechanisms to prevent hallucinated product attributes. Exposed the system via a FastAPI service with caching and a conversational web interface.

Technologies: Python, FastAPI, SQLite, ChromaDB, sentence-transformers, Pydantic, agentic AI, RAG, hybrid retrieval

Education

- 2009 – 2015 **PhD in Automatic Control, KTH Royal Institute of Technology, Stockholm**
Dissertation: Architectures and Performance Analysis of Wireless Control Systems
Advisor: Prof. Mikael Johansson
- 2007 – 2009 **MSc in Mechatronics Engineering, Istanbul Technical University, Istanbul**
GPA: 3.79/4.00
- 2004 – 2009 **BSc in Control Engineering, Istanbul Technical University, Istanbul**
GPA: 3.52/4.00
- 2003 – 2007 **BSc in Mechanical Engineering, Istanbul Technical University, Istanbul**
GPA: 3.64/4.00, 1st among 155+

Technical Skills

AI/ML Frameworks & Paradigms

- **Agentic AI systems:** autonomous decision-making, tool use, planning, closed-loop control
- **Reinforcement learning & control:** policy optimization, sequential decision-making, multi-objective RL
- **LLM-integrated architectures:** reasoning, tool calling
- **Retrieval-Augmented Generation (RAG):** vector search, embedding-based retrieval, context grounding
- **Model compression:** policy distillation, domain-specific and lightweight models

Programming

- **Python (expert)**: PyTorch, NumPy, Scikit-learn, Hugging Face
- **Go (basic)**: microservices, concurrency

Systems & Infrastructure

- Linux, Docker, Kubernetes, GitLab CI/CD
- Internal/private cloud platforms, AWS
- HPC: LSF (advanced), SLURM (intermediate)

Awards & Certifications

- 2025, 2022 **Key Contributor Award**, Ericsson
- 2023 **Leaders' Core Curriculum Program**, Ericsson
- 2023 **High-Impact Decision Making Certification**, Mentora Institute

Publications & Patents

- Author of **25+ peer-reviewed papers** in machine learning, control, and optimization.
- Inventor on **15+ patents** in AI/ML, network automation, and autonomous systems.
- Full list available: [Google Scholar](#)